sAIs Project Charter

Smart Autonomous Intelligent System

Tagline: Your parking attendant, reimagined.

Document Information

Document Information	Details
Project Name	sAIs
Full Name	Smart Autonomous Intelligent System
Project Category	Smart Mobility & Autonomous Parking
Project Type	Engineering System Design & Proposal
Primary Deployment	Indoor Shopping Mall Parking Facility
Team Size	6 Members
Document Version	3.0
Project Status	Baseline – Planning Phase

1. Executive Summary

sAIs (Smart Autonomous Intelligent System) is a proposed intelligent parking ecosystem designed to transform conventional indoor shopping mall parking facilities into automated, connected, and data-driven environments.

The system combines autonomous robotic valet technology, IoT-based parking monitoring, intelligent space allocation, mobile application services, and AI-powered demand forecasting.

Unlike conventional parking systems, sAIs introduces an autonomous vehicle-handling approach in which specialized mobile robots collect, transport, and retrieve customers' vehicles without requiring customers to navigate the parking facility or manually park their vehicles.

Customers interact with the system through a dedicated mobile application, allowing them to register, check parking availability, reserve services, and request vehicle retrieval.

The platform also incorporates predictive analytics to help customers identify less congested arrival periods while enabling facility operators to optimize parking utilization and operational efficiency.

The proposed system takes inspiration from commercially available autonomous parking technologies, particularly HL Robotics' PARKIE platform, while extending the concept through integrated digital services and intelligent parking management.

The initial project focuses exclusively on an indoor parking environment, with a hybrid robotic coordination architecture combining centralized fleet management and autonomous robot-level decision-making.

2. Project Background and Problem Statement

Shopping mall parking facilities frequently experience operational challenges, particularly during peak hours and high-demand periods.

These challenges include:

- Drivers spending considerable time searching for available parking spaces.
- Internal traffic congestion caused by vehicles circulating through parking areas.
- Inefficient utilization of available parking capacity.
- Limited real-time visibility of parking availability.
- Difficulty managing premium parking reservations.
- Dependence on manual parking operations.
- Poor integration between parking services and customer-facing digital platforms.

Traditional parking management solutions primarily focus on identifying available spaces rather than automating the complete vehicle parking and retrieval process.

sAIs addresses this limitation by combining intelligent parking management with autonomous robotic vehicle transportation.

3. Project Objectives

The primary objectives of sAIs are:

1. Develop an intelligent parking management architecture capable of monitoring parking occupancy in real time.
2. Introduce autonomous robotic valet technology to automate vehicle transportation, storage, and retrieval.
3. Implement an intelligent parking allocation mechanism that considers vehicle requirements, availability, and customer preferences.
4. Provide a mobile application for customer registration, parking reservations, vehicle monitoring, and service requests.
5. Integrate AI-based analytics to forecast parking demand and recommend optimal arrival periods.
6. Improve parking facility utilization through automated vehicle storage and intelligent allocation.
7. Develop a scalable system architecture that can accommodate future services and additional functionality.

4. Project Scope

4.1 In Scope

A. Intelligent Parking Management

- Real-time parking occupancy monitoring.
- Intelligent parking space allocation.
- Vehicle classification and parking requirements.
- Parking availability visualization.
- Reservation management.
- Parking facility monitoring.

B. Autonomous Robotic Valet

- Autonomous mobile robots for vehicle transportation.
- Automated vehicle pickup and retrieval.
- Hybrid multi-robot coordination.
- Robot localization and navigation.
- Obstacle detection and collision avoidance.
- Robot status monitoring and fleet supervision.
- Automated charging management.

C. Mobile Application

- User registration and authentication.
- Real-time parking availability.
- VIP parking reservations.
- Vehicle retrieval requests.
- Vehicle location and parking-status monitoring.
- Service booking and notifications.
- Digital payment integration.

D. AI and Predictive Analytics

- Historical and real-time parking data analysis.
- Parking occupancy forecasting.
- Demand pattern recognition.
- Recommended arrival periods.
- Potential dynamic pricing optimization.

E. Future Expansion Opportunities

The following services may be integrated in later project phases:

- EV charging station integration.
- Automated vehicle washing stations.
- Additional vehicle monitoring and security services.

These services are not part of the initial core implementation scope.

4.2 Out of Scope

The initial project proposal does not include:

- Manufacturing a commercial-grade autonomous parking robot.
- Constructing or modifying a complete commercial parking facility.
- Deploying a production-ready mobile application.
- Training and deploying a fully operational AI prediction model.
- Commercial operation or maintenance of a real-world facility.
- Outdoor parking deployment.
- Implementing EV charging or automated vehicle washing as core subsystems.

These activities may be considered during subsequent implementation phases.

5. Technology Benchmark and Preliminary Design Basis

The project will reference commercially available autonomous parking technologies to establish a technically realistic foundation.

5.1 Autonomous Parking Robot References

Primary Technology Reference: HL Robotics PARKIE

PARKIE is an autonomous mobile robot designed for indoor parking applications.

Its documented capabilities include:

- Autonomous navigation.
- Obstacle and pathway detection.
- Vehicle wheel-position identification.
- Mechanical vehicle lifting and transportation.
- Integration with mobile applications and external parking management systems through APIs.

PARKIE will serve as the primary technological benchmark for the proposed indoor robotic valet subsystem.

Comparative Reference: HL Robotics STAN

STAN is an autonomous parking robot designed primarily for outdoor environments.

Its documented capabilities include:

- Autonomous vehicle transportation.
- Outdoor operation under varying weather conditions.
- Vehicle payload capacity of up to 3 tonnes.
- Automated charging.
- Integration with digital-twin and robot-supervision software.

Although STAN is not intended for the initial indoor deployment, it may provide useful comparative insights into autonomous parking technology and future scalability.

These platforms are technology benchmarks, not confirmed procurement selections. Final technology choices will depend on technical feasibility, compatibility, availability, and budget evaluation.

5.2 Preliminary Technology Selection

System Component	Proposed Technology / Reference
Indoor parking robots	HL Robotics PARKIE as the primary benchmark
Robot navigation	LiDAR-based localization and SLAM
Vehicle and obstacle detection	LiDAR, cameras, and proximity sensors
Parking occupancy monitoring	Ultrasonic or camera-based detection
Robot coordination	Hybrid fleet management architecture
Communication	Wi-Fi / MQTT / REST APIs
Application	Cross-platform mobile application
AI analytics	Historical data analysis and predictive models
Data management	Centralized database and cloud services

The listed technologies represent a preliminary design basis. Final component selection will depend on technical feasibility, system requirements, compatibility, and budget evaluation.

5.3 Proposed System Architecture

The system will use a hybrid robotic coordination architecture.

It will combine centralized management with autonomous robot-level control.

Centralized Fleet Management Responsibilities:

- Assigning parking and retrieval tasks to available robots.
- Managing fleet status and robot availability.
- Coordinating task priorities.
- Monitoring robot operations.
- Managing overall parking allocation.
- Resolving fleet-level scheduling and coordination requirements.

Individual Robot Responsibilities:

- Local navigation and localization.
- Obstacle detection and avoidance.
- Immediate movement and safety decisions.
- Executing assigned transportation tasks.
- Reporting operational status to the fleet manager.

This architecture provides centralized operational oversight while allowing robots to make immediate local decisions without depending on the central controller for every movement.

It should not be described as a fully decentralized swarm. It is a hybrid multi-robot fleet with centralized task management and autonomous local control.

The wider system consists of five main subsystems:

1. **Parking Management System:** Monitors occupancy and manages parking allocation.
2. **Robotic Fleet Management System:** Assigns tasks and supervises autonomous robot operations.
3. **Mobile Application:** Provides customer interaction and service management.
4. **AI Analytics Engine:** Processes operational data and predicts parking demand.
5. **Smart Services Management:** Provides the architecture for integrating future services.

6. Operational Concept

The proposed customer journey is as follows:

Stage 1 – Pre-Arrival

The customer opens the sAIs mobile application to register, check parking availability, view predicted occupancy, and explore available reservations.

Stage 2 – Arrival and Vehicle Handover

The customer arrives at a designated indoor vehicle drop-off area and selects the autonomous valet service through the application.

Stage 3 – Autonomous Vehicle Transportation

The fleet manager assigns an available robot to the task. The robot independently performs local navigation, obstacle avoidance, and vehicle handling before transporting the vehicle to its allocated parking location.

Stage 4 – Parking Monitoring

The customer can monitor the vehicle's parking status and request available services through the application.

Stage 5 – Vehicle Retrieval

Upon departure, the customer requests vehicle retrieval. The fleet manager assigns an available robot, which autonomously transports the vehicle to a designated pickup area.

7. Project Stakeholders

Stakeholder	Responsibility / Interest
Project Team	System design, planning, integration, and documentation
Project Supervisor	Project guidance, review, and evaluation
Shopping Mall Management	Facility operation, efficiency, and profitability
Parking Customers	System usability, convenience, and vehicle safety
Technology Suppliers	Hardware and software integration
Maintenance Personnel	Equipment reliability and maintenance
Facility Security	Operational safety and access control

8. Major Project Deliverables

The project consists of the following eleven deliverables:

1. Project Charter.
2. Project Plan.
3. Resource Allocation Plan.
4. Risk Management Plan.
5. Project Schedule.
6. Communication Plan.
7. Quality Assurance Plan.
8. Budget Management Plan.
9. Industry Case Study.
10. Final Project Report.
11. Technical Presentation.

9. Project Constraints

- Limited project preparation time.
- Six-member project team.
- Financial and procurement limitations.
- Complexity of autonomous vehicle transportation.
- Integration challenges between robotic and digital systems.
- Safety requirements associated with vehicle handling.
- Dependence on reliable communication infrastructure.
- Limited access to real-world operational data.
- Dependence on the technical capabilities and operating requirements of the selected robotic platform.

10. Project Assumptions

- The primary deployment environment is an indoor shopping mall parking facility.
- The facility contains designated vehicle drop-off and pickup areas.
- The indoor environment can accommodate autonomous robotic operations.
- Vehicles meet the operating requirements of the selected robotic platform.
- The robotic fleet operates within predefined controlled zones.
- Customers interact with the system through a mobile application.
- A central fleet manager coordinates task allocation and fleet supervision.
- Individual robots retain local navigation and immediate safety capabilities.
- Historical and real-time operational data can be collected for analytical purposes.
- The project will use commercially available technology as a design reference rather than assume that a particular commercial robot will be purchased.

11. Project Success Criteria

The proposed project will be considered successful at the design and planning level when:

- A complete and logically consistent system architecture is established.
- The indoor robotic valet subsystem is supported by commercially available technology references.
- The hybrid robot coordination architecture is clearly defined.
- Parking allocation and reservation workflows are established.
- The mobile application concept covers the essential customer journey.
- The AI analytics subsystem has defined inputs, outputs, and objectives.
- Technical, operational, and safety risks are identified.
- Required resources and financial requirements are estimated.
- All project management documents are completed and reviewed.
- The final proposal demonstrates technical feasibility and a clear implementation strategy.

Quantitative performance targets will be established during the requirements and feasibility analysis stages.

12. Preliminary Project Risks

The following high-level risks have been identified:

- Robotic navigation or localization failure.
- Vehicle lifting mechanism malfunction.
- Robot-to-robot collisions.
- Communication interruptions between robots and the fleet manager.
- Incorrect parking allocation.
- Inaccurate AI predictions.
- Cybersecurity and customer-data privacy concerns.
- High initial implementation and maintenance costs.
- Incompatibility between the selected robot platform and facility layout.

Detailed risk evaluation, prioritization, and mitigation measures will be documented in the Risk Management Plan.

13. Preliminary Budget Considerations

The project budget will distinguish between:

Capital Expenditure (CAPEX):

- Autonomous parking robots.
- Parking infrastructure modifications.
- Occupancy detection equipment.
- Communication infrastructure.
- Software development and integration.

Operational Expenditure (OPEX):

- Electricity and charging.
- Robot maintenance.
- Software subscriptions.
- Network services.
- Operational personnel.
- Replacement components.

A detailed cost estimation and financial feasibility assessment will be developed in the Budget Management Plan.

14. Project Team Structure and Responsibilities

The six-member team will operate with clearly assigned responsibilities. Each member is accountable for the deliverables assigned to them and will contribute to the overall project integration.

Team Member	Assigned Deliverables	Primary Role	Key Responsibilities
Mohamed Yasser	Project Charter, Communication Plan	Project Coordination & Communication	Establishing the project baseline, defining project objectives and scope, coordinating communication, maintaining stakeholder updates, and supporting overall document integration.
Farha	Risk Management Plan, Industry Case Study	Risk & Research Analyst	Identifying and evaluating project risks, proposing mitigation strategies, researching relevant industry implementations, and extracting lessons applicable to sAIs.
Omar	Project Plan, Quality Assurance Plan	Project Planning & Quality Lead	Developing the WBS, defining project activities and dependencies, establishing planning methods, defining quality standards, and preparing verification and review procedures.
Salma	Budget Management Plan	Budget & Cost Analyst	Identifying cost categories, preparing preliminary cost estimates, distinguishing CAPEX from OPEX, and evaluating financial feasibility and budget assumptions.
Mohamed Hany	Resource Allocation Plan	Resource Planning Coordinator	Identifying required human, hardware, software, and infrastructure resources, allocating resources to activities, and identifying resource constraints.
Peter	Project Schedule	Scheduling Coordinator	Developing the project timeline, defining task durations and dependencies, establishing milestones, and monitoring schedule progress.

14.1 Accountability and Collaboration

- Each deliverable has one designated owner responsible for its completion and initial quality.
- Deliverable owners are expected to coordinate with other members whenever their work depends on information from another document.
- The Project Charter, Project Plan, and Project Schedule establish the foundation for the remaining planning documents.
- The Project Plan and Project Schedule must remain consistent with the approved scope and team allocation.
- The team will review the combined documents to identify contradictions, duplicated work, and missing information.
- All members will contribute to the Final Project Report and Technical Presentation, even if they are not the primary owners of those deliverables.

14.2 Project Coordination

The project will be coordinated collectively, with Mohamed Yasser responsible for maintaining the project baseline and communication structure.

This coordination responsibility does not replace the accountability of individual deliverable owners.

Major scope changes, significant resource changes, and modifications to project objectives must be discussed and agreed upon by the team and submitted to the project supervisor when approval is required.

15. Project Governance and Approval

The project will be managed through a shared responsibility model.

- **Project Coordination:** Maintains project alignment, communication, and baseline consistency.
- **Deliverable Ownership:** Each member is accountable for their assigned documents.
- **Team Review:** Members review and provide feedback on interconnected deliverables.
- **Supervisor Review:** The project supervisor provides guidance and evaluates the proposed project.

Role	Name	Approval
Project Coordination	Mohamed Yasser	Pending
Risk & Research Analyst	Farha	Pending
Project Planning & Quality Lead	Omar	Pending
Budget & Cost Analyst	Salma	Pending
Resource Planning Coordinator	Mohamed Hany	Pending
Scheduling Coordinator	Peter	Pending
Project Supervisor	TBD	Pending

16. Baseline Declaration

This charter establishes the initial approved-by-team planning baseline for the sAIs project.

The baseline defines:

- **Project Identity:** sAIs – Smart Autonomous Intelligent System.
- **Deployment Environment:** Indoor shopping mall parking facility.
- **Robotic Coordination:** Hybrid centralized fleet management and autonomous local robot control.
- **Core Scope:** Intelligent parking management, autonomous robotic valet, mobile application, and AI-powered analytics.
- **Team Structure:** Six members with designated deliverable ownership and responsibilities.
- **Project Governance:** Shared team accountability with designated project coordination.

Any significant change to the project scope, deployment environment, system architecture, or team responsibilities must be reviewed and documented before being incorporated into the baseline.

Document Status: Project Charter – Baseline for Team and Supervisor Approval

Project Identity: sAIs – Your parking attendant, reimagined.